



**The CENTRE for EDUCATION
in MATHEMATICS and COMPUTING**

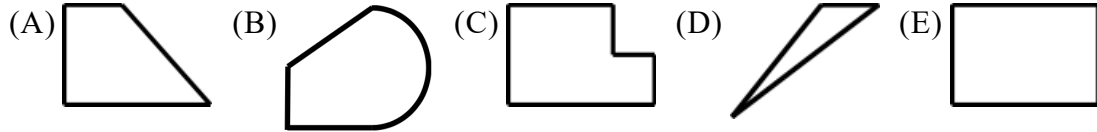
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**2024 Gauss Grade 8
Problems**

1. How many 5 cent coins are needed to make 25 cents?

- (A) 1 (B) 2 (C) 3 (D) 4 (E) 5
-

2. Which of the following shapes has a vertical line of symmetry?



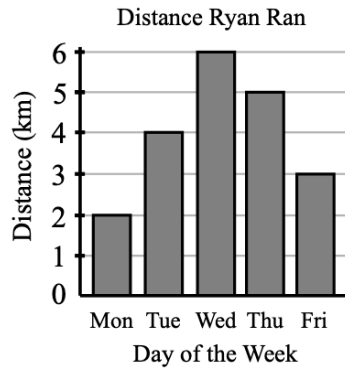
3. Which of the following numbers is largest?

- (A) 0.58 (B) 1.32 (C) 0.97 (D) 1.03 (E) 0.12
-

4. 50% of n is 2024. The value of n is

- (A) 2074 (B) 24 (C) 50 (D) 4048 (E) 4042
-

5. Ryan recorded the distance, in kilometres, that he ran on each day from Monday to Friday, as shown.



The total distance that Ryan ran over the five days is

- (A) 14 km (B) 16 km (C) 18 km (D) 20 km (E) 22 km
-

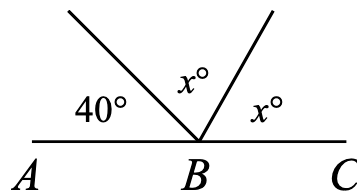
6. When the number 11 is increased by 2 and the result is then multiplied by 3, the final result is

- (A) 35 (B) 39 (C) 28 (D) 25 (E) 363
-

7. The value of a that satisfies the equation $15 + a = 10$ is

- (A) -10 (B) -5 (C) 0 (D) 5 (E) 10
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8. In the diagram, $\angle ABC$ is a straight angle.

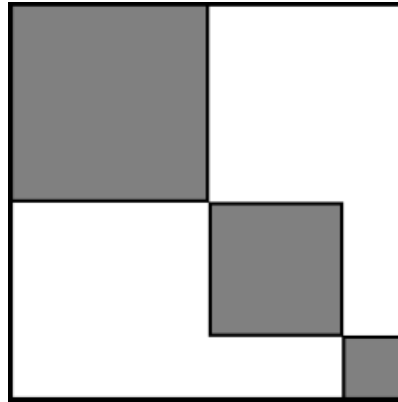


The value of x is

- (A) 80 (B) 65 (C) 75 (D) 70 (E) 60
-

9. In a drawer, the ratio of the number of spoons to the number of forks is $1 : 2$. The total number of spoons and forks in the drawer *cannot* be equal to
- (A) 12 (B) 6 (C) 18 (D) 10 (E) 3
-

10. In the diagram, a square with side length 6 is partially shaded.



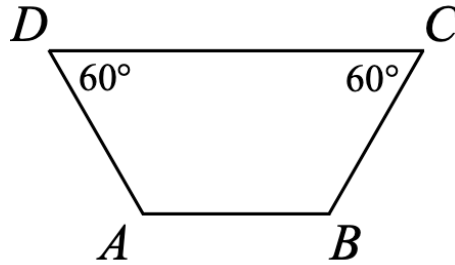
The largest shaded region is a square with side length 3. The other two shaded regions are squares with side lengths 2 and 1. What is the total area of the unshaded region?

- (A) 12 (B) 18 (C) 22 (D) 24 (E) 30
-

11. In the sequence $1, 3, 4, 7, \dots$, each number beginning with the 4 is the sum of the two numbers before it. This means that the next number in the sequence is $4 + 7 = 11$. The smallest number greater than 100 that appears in the sequence is
- (A) 133 (B) 199 (C) 113 (D) 101 (E) 123
-

12. The number 385 has three prime factors. The sum of these prime factors is
- (A) 21 (B) 26 (C) 25 (D) 23 (E) 22
-

13. Trapezoid $ABCD$ can be divided into three equilateral triangles.



If the perimeter of the trapezoid is equal to 840cm , what is the length of AB ?

- (A) 120 cm (B) 140 cm (C) 168 cm (D) 25 cm (E) 210 cm
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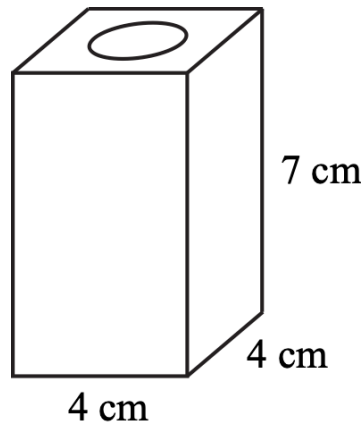
14. A container of ice cream can make 6 cones or it can make 4 sundaes. If 5 such containers of ice cream are used to make 12 cones, what is the greatest number of sundaes that can be made with the ice cream that remains?

- (A) 4 (B) 8 (C) 12 (D) 16 (E) 20
-

15. When a positive integer n is divided by 10, the remainder is 8. When n is divided by 5, the remainder is

- (A) 0 (B) 1 (C) 2 (D) 3 (E) 4
-

16. A block of wood in the shape of a rectangular prism has length 4 cm, width 4 cm, and height 7 cm. A cylindrical hole with radius 1 cm is drilled through the centre, as shown.

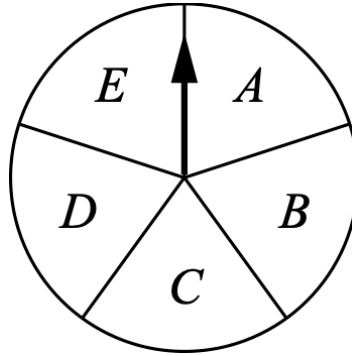


To the nearest cm^3 , what is the volume of the block of wood after the hole is drilled? (Note: The volume of a cylinder with radius r and height h is $\pi r^2 h$.)

- (A) 90 cm^3 (B) 122 cm^3 (C) 106 cm^3 (D) 84 cm^3 (E) 92 cm^3
-

17. The Gaussbot factory assembles robots. Each robot comes in one of three colours: red, blue, or green. Each robot also has a number stamped on its head: 1, 2, 3, or 4. The n th robot assembled is the first robot to have the same colour and the same number as a previously assembled robot. What is the greatest possible value of n ?
- (A) 11 (B) 12 (C) 13 (D) 7 (E) 8
-

18. A circular spinner is divided into five equal sections. An arrow is attached to the centre of the spinner and is positioned as shown.



The arrow is spun clockwise, and it stops in the section labelled *D*. Which of the following could have been the angle of rotation?

- (A) 530° (B) 550° (C) 630° (D) 675° (E) 700°
-

19. Three *different* integers in a list have a mean (average) of 50 and a range of 14. What is the smallest possible integer in the list?

- (A) 40 (B) 43 (C) 39 (D) 42 (E) 41
-

20. Kiran has a box containing three different types of fruit: apples, pears, and bananas. In the box, 21 pieces of fruit are not apples, 25 pieces of fruit are not pears, and 28 pieces of fruit are not bananas. How many pieces of fruit are in the box?

- (A) 53 (B) 32 (C) 46 (D) 37 (E) 51
-

21. The prime factorization of $6 \times 5 \times 4 \times 3 \times 2 \times 1$ can be written in the form $2^a \times 3^b \times 5^c$. The value of $a + b + c$ is

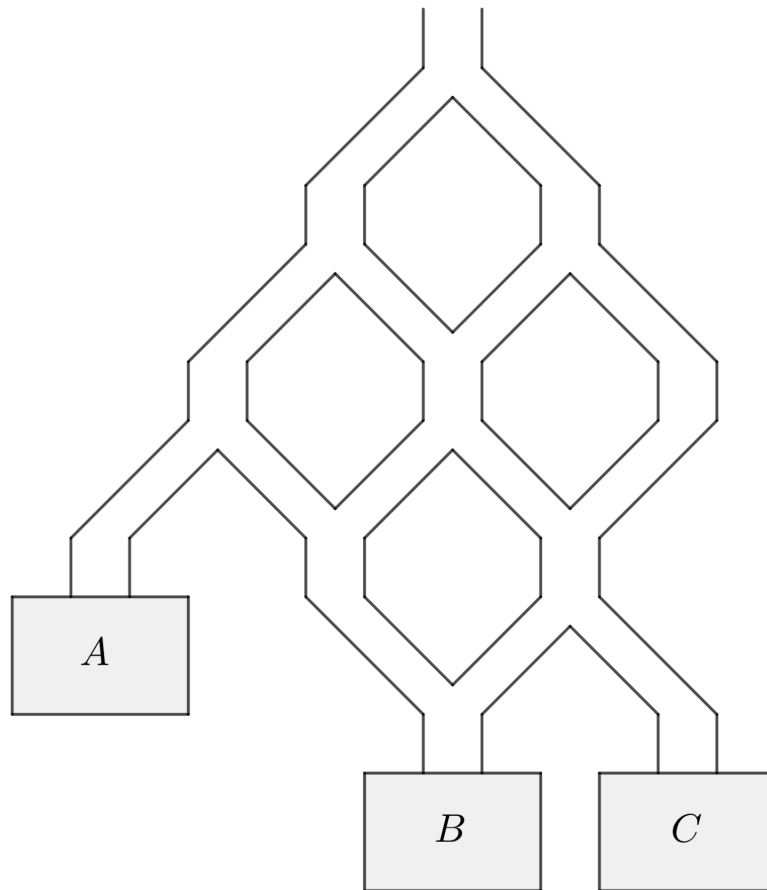
- (A) 6 (B) 7 (C) 9 (D) 5 (E) 8
-

22. The ratio of the number of quarters (worth \$0.25 each) to the number of dimes (worth \$0.10 each) to the number of nickels (worth \$0.05 each) in a jar is $9 : 3 : 2$. If the total value of the quarters and dimes is \$17.85, what is the total value of the nickels?

- (A) \$0.45 (B) \$0.50 (C) \$0.70 (D) \$0.35 (E) \$0.55
-

23. Five *different* integers, each greater than 0, have a sum of 264. The greatest common divisor of these five positive integers is d . What is the sum of the digits of the largest possible value of d ?
- (A) 4 (B) 3 (C) 8 (D) 2 (E) 6
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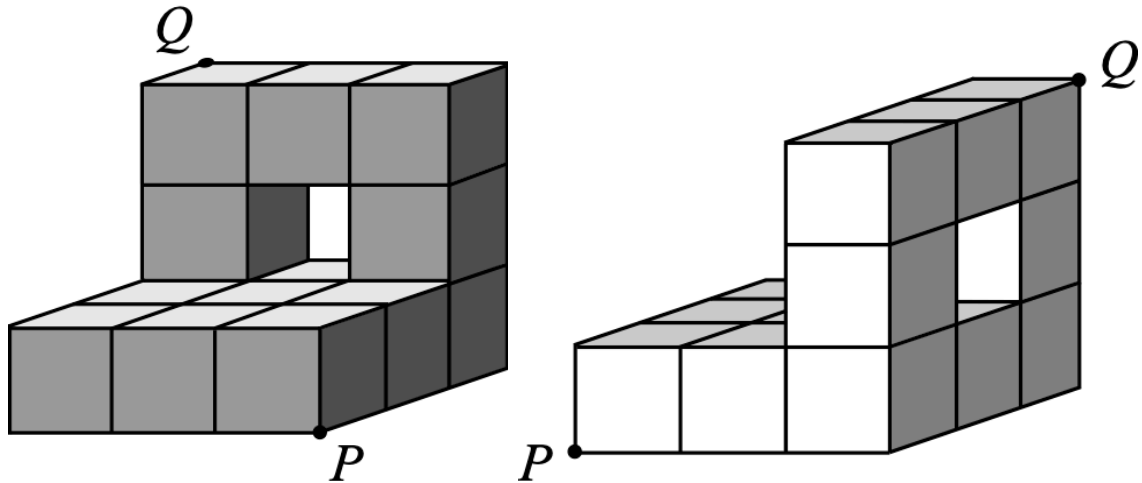
24. A network of pathways lead from a single opening to three bins, labelled A , B , C as shown. If a ball is dropped into the opening, it will follow a path and land in one of the bins. Every time a path splits, it is equally likely for the ball to follow either of the downward paths.



Ellen drops two balls, one after the other, into the opening. What is the probability that the two balls land in different bins?

- (A) $\frac{17}{32}$ (B) $\frac{27}{50}$ (C) $\frac{25}{64}$ (D) $\frac{1}{3}$ (E) $\frac{15}{32}$
-

25. A figure is constructed using fourteen $1 \times 1 \times 1$ cubes. Nine of the $1 \times 1 \times 1$ cubes are used to make the bottom layer and five additional $1 \times 1 \times 1$ cubes are positioned on top of the bottom layer. The figure is shown from two different perspectives.



First perspective

9 cubes form a 3 by 3 base layer of the figure, arranged into a front row, a middle row, and a back row.

5 additional cubes are placed above the back row forming a shape like a rectangular arch: 2 cubes are stacked on top of both the leftmost cube and the rightmost cube, forming a middle layer and a top layer, and 1 cube is placed between the 2 top cubes completing a row of 3 cubes in the top layer.

Points P and Q mark opposite corners of the figure: P is located at the bottom-right corner of the front face and Q is located at the top-left corner of the back face.

Second perspective

First perspective rotated ninety degrees so that the right face in the first view becomes the front face in the second view. In this view, P is located at the bottom-left corner of the front face and Q is located at the top-right corner of the back face.

An ant begins at P and walks a distance d on the surface of the figure to arrive at Q . The smallest possible value of d is closest to

- (A) 6.43 (B) 6.40 (C) 6.71 (D) 6.66 (E) 6.48
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